**MCQ – Lecture 1**

1. Software Testing is mainly concerned with:

- a) Writing source code
- b) Executing programs to find defects
- c) Designing system architecture
- d) Deploying applications
  **Answer: b**

2. Quality Assurance in software engineering focuses on:

- a) Detecting bugs after release
- b) Fixing runtime errors
- c) Improving processes to prevent defects
- d) Writing test cases only
  **Answer: c**

3. The relationship between Software Testing and Quality Assurance is that:

- a) They are identical concepts
- b) Testing replaces quality assurance
- c) Testing is part of quality assurance activities
- d) Quality assurance is part of testing
  **Answer: c**

4. Common coding mistakes are:

- a) Errors caused by hardware failure
- b) Rare programming errors
- c) Frequently occurring programming errors
- d) Errors found only during acceptance testing
  **Answer: c**

5. Debugging is the process of:

- a) Executing test cases
- b) Measuring software performance
- c) Locating and fixing bugs

- d) Writing documentation
  **Answer: c**

6. Debugging is performed:

- a) Before coding
- b) After detecting errors
- c) Instead of testing
- d) Without executing the program
  **Answer: b**

7. Capability Maturity Model (CMM) is used to:

- a) Measure software size
- b) Evaluate software development processes
- c) Detect runtime errors
- d) Design user interfaces
  **Answer: b**

8. CMM focuses on improving:

- a) Source code syntax
- b) Hardware performance
- c) Software development processes
- d) Programming languages
  **Answer: c**

9. Improving software processes using CMM aims to:

- a) Reduce testing cost only
- b) Eliminate defects completely
- c) Positively affect software quality
- d) Increase system complexity
  **Answer: c**

10. CMM evaluates processes used for:

- a) Executing software projects
- b) Writing test scripts

- c) Managing databases
- d) Designing UI screens
  **Answer: a**

11. Software quality according to CMM is mainly improved by:

- a) Hiring more testers
- b) Improving development processes
- c) Increasing execution speed
- d) Using advanced hardware
  **Answer: b**

12. CMM is concerned with:

- a) Final software product only
- b) Processes that produce the software
- c) Testing tools selection
- d) Debugging techniques
  **Answer: b**

13. An error is:

- a) A failure observed by the user
- b) A mistake made by a human
- c) A system crash
- d) A defect found after delivery
  **Answer: b**

14. A defect is:

- a) A human mistake
- b) Incorrect software behavior during execution
- c) A flaw in software caused by an error
- d) A hardware problem
  **Answer: c**

15. A failure occurs when:

- a) An error is made by the developer

- b) A defect exists in the code
- c) The system behaves incorrectly at runtime
- d) The code is reviewed
  **Answer: c**

16. Testing helps in:

- a) Proving the software is bug-free
- b) Increasing software cost
- c) Reducing risk of failures
- d) Eliminating debugging
  **Answer: c**

17. The main objective of testing is to:

- a) Show software works correctly
- b) Find defects
- c) Improve performance
- d) Write documentation
  **Answer: b**

18. Verification answers the question:

- a) Are we building the right product?
- b) Are we building the product right?
- c) Does the system meet user needs?
- d) Is the system usable?
  **Answer: b**

19. Validation answers the question:

- a) Are we building the product right?
- b) Is the code optimized?
- c) Are we building the right product?
- d) Is the system error-free?
  **Answer: c**

20. Verification activities mainly involve:

- a) Executing the program
- b) Reviewing documents and code
- c) User acceptance testing
- d) Performance testing
  **Answer: b**

21. Validation is mainly performed by:

- a) Developers only
- b) Testers only
- c) End users
- d) Configuration managers
  **Answer: c**

22. Testing differs from debugging because testing:

- a) Fixes defects
- b) Prevents errors
- c) Finds defects
- d) Improves performance
  **Answer: c**

23. Debugging is performed:

- a) Before testing
- b) During requirements analysis
- c) After defects are detected
- d) Instead of verification
  **Answer: c**

24. Static testing is performed:

- a) By executing the program
- b) Without executing the program
- c) Only by end users
- d) After deployment
  **Answer: b**

25. Dynamic testing requires:

- a) Code inspection
- b) Requirement review
- c) Program execution
- d) Design analysis
  **Answer: c**

26. Which of the following is a static testing technique?

- a) Unit testing
- b) Integration testing
- c) Code review
- d) System testing
  **Answer: c**

27. Which of the following is a dynamic testing technique?

- a) Walkthrough
- b) Inspection
- c) Code reading
- d) Unit testing
  **Answer: d**

28. Unit testing focuses on:

- a) Entire system behavior
- b) Interaction between modules
- c) Smallest testable unit
- d) User requirements
  **Answer: c**

29. Integration testing focuses on:

- a) Individual functions
- b) Combined modules interaction
- c) User acceptance

- d) System performance
  **Answer: b**

30. System testing is concerned with:

- a) Individual methods
- b) Module interfaces
- c) Complete integrated system
- d) Source code inspection
  **Answer: c**

31. Acceptance testing is mainly used to:

- a) Detect coding errors
- b) Validate system with user requirements
- c) Optimize performance
- d) Debug runtime errors
  **Answer: b**

32. The first level in CMM is:

- a) Repeatable
- b) Defined
- c) Initial
- d) Optimizing
  **Answer: c**

33. The Initial level in CMM is characterized by:

- a) Well-defined processes
- b) Quantitatively managed processes
- c) Ad-hoc and chaotic processes
- d) Continuous improvement
  **Answer: c**

34. In CMM Level 1, success mainly depends on:

- a) Process maturity
- b) Individual effort

- c) Automated tools
- d) Documentation quality
  **Answer: b**

35. The Repeatable level in CMM mainly focuses on:

- a) Process optimization
- b) Cost, schedule, and functionality
- c) Statistical control
- d) Innovation
  **Answer: b**

36. In CMM Level 2, project management practices are:

- a) Undefined
- b) Informal
- c) Planned and tracked
- d) Ignored
  **Answer: c**

37. The Defined level in CMM indicates that processes are:

- a) Ad-hoc
- b) Individually dependent
- c) Documented and standardized
- d) Unmeasured
  **Answer: c**

38. Which CMM level emphasizes organization-wide standards?

- a) Initial
- b) Repeatable
- c) Defined
- d) Managed
  **Answer: c**

39. Quantitatively Managed level focuses on:

- a) Code reviews

- b) Documentation
- c) Statistical and quantitative techniques
- d) Individual skills
  **Answer: c**

40. At which CMM level are software processes measured and controlled?

- a) Level 2
- b) Level 3
- c) Level 4
- d) Level 5
  **Answer: c**

41. The Optimizing level in CMM is mainly concerned with:

- a) Process definition
- b) Quantitative control
- c) Continuous process improvement
- d) Basic project management
  **Answer: c**

42. Continuous improvement in CMM is achieved by:

- a) Eliminating testing
- b) Preventing defects
- c) Increasing manpower
- d) Reducing documentation
  **Answer: b**

43. Key Process Areas (KPAs) are used to:

- a) Measure code size
- b) Define process goals and practices
- c) Replace testing
- d) Perform debugging
  **Answer: b**

44. KPAs are associated with:

- a) Programming languages
- b) Hardware platforms
- c) Specific CMM maturity levels
- d) Testing tools
  **Answer: c**

45. KPAs help organizations to:

- a) Improve software processes
- b) Increase system failures
- c) Reduce code readability
- d) Eliminate management activities
  **Answer: a**

46. An error discovered before software delivery is classified as:

- a) Failure
- b) Defect
- c) Error
- d) Crash
  **Answer: c**

47. A defect is discovered:

- a) Before testing
- b) Before coding
- c) After delivery to the end user
- d) During requirements analysis
  **Answer: c**

48. Software measurement is used to:

- a) Write source code
- b) Quantify attributes of software
- c) Debug programs
- d) Replace testing
  **Answer: b**

49. Direct software measures include:

- a) Quality and complexity
- b) Functionality and usability
- c) Lines of Code and effort
- d) Maintainability
  **Answer: c**

50. Indirect software measures include:

- a) Lines of Code
- b) Cost
- c) Effort
- d) Quality
  **Answer: d**

51. Lines of Code (LOC) is considered:

- a) An indirect measure
- b) A direct measure
- c) A qualitative metric
- d) A usability metric
  **Answer: b**

52. Size-oriented metrics use:

- a) Function Points
- b) Lines of Code as normalization value
- c) Number of users
- d) Execution time
  **Answer: b**

53. Errors per KLOC is an example of:

- a) Function-oriented metric
- b) Size-oriented metric
- c) Usability metric

- d) Performance metric
  **Answer: b**

54. One major criticism of LOC is that it is:

- a) Language independent
- b) Easy to estimate
- c) Language dependent
- d) User oriented
  **Answer: c**

55. Function Point metrics are:

- a) Language dependent
- b) Based on LOC only
- c) Language independent
- d) Hardware dependent
  **Answer: c**

56. Function Points are based on:

- a) Code syntax
- b) Information domain and complexity
- c) Execution speed
- d) Number of programmers
  **Answer: b**

57. Object-oriented metrics may include:

- a) Number of classes and scenarios
- b) Lines of Code only
- c) Disk usage
- d) CPU utilization
  **Answer: a**

58. Use-case oriented metrics are:

- a) Dependent on programming language
- b) Independent of programming language

- c) Based on LOC
- d) Based on compiler type
  **Answer: b**

59. A major issue with use-case oriented metrics is:

- a) High cost
- b) Lack of standard size for use cases
- c) Language dependency
- d) Complex calculation
  **Answer: b**

60. Software quality metrics mainly focus on:

- a) Code formatting
- b) Number of users
- c) Errors and defects
- d) Hardware speed
  **Answer: c**

61. Correctness is usually measured by:

- a) Response time
- b) Defects per KLOC
- c) CPU utilization
- d) Throughput
  **Answer: b**

62. Maintainability can be measured using:

- a) MTTC (Mean Time To Change)
- b) Bandwidth
- c) Disk usage
- d) Latency
  **Answer: a**

63. Defect Removal Efficiency (DRE) is calculated as:

- a) D / (E + D)

- b) E / (E + D)
- c) E / D
- d) D / E
**Answer: b**

64. A DRE value close to 1 indicates:

- a) Poor quality
- b) Many remaining defects
- c) High defect removal efficiency
- d) No testing effort
**Answer: c**

65. A common programming mistake related to variables is:

- a) Using meaningful names
- b) Using uninitialized variables
- c) Declaring variables before use
- d) Limiting variable scope
**Answer: b**

66. Using an uninitialized variable may lead to:

- a) Compile-time warning only
- b) Predictable output
- c) Unexpected or incorrect results
- d) Better performance
**Answer: c**

67. A variable scope determines:

- a) Variable data type
- b) Variable lifetime and visibility
- c) Variable memory size
- d) Variable execution speed
**Answer: b**

68. Using global variables excessively is considered a mistake because it:

- a) Improves readability
- b) Simplifies debugging
- c) Increases coupling and errors
- d) Reduces memory usage
  **Answer: c**

69. One common mistake in conditional statements is:

- a) Using proper logical operators
- b) Forgetting else conditions
- c) Writing clear conditions
- d) Using parentheses
  **Answer: b**

70. A missing break statement in a switch-case structure may cause:

- a) Compilation error
- b) Infinite loop
- c) Fall-through execution
- d) Memory leak
  **Answer: c**

71. Using == instead of = in conditions may result in:

- a) Better comparison
- b) Assignment instead of comparison
- c) Logical error
- d) Syntax error always
  **Answer: c**

72. Infinite loops are usually caused by:

- a) Correct loop conditions
- b) Loop variables updated properly
- c) Incorrect or missing termination conditions
- d) Compiler optimizations
  **Answer: c**

73. An off-by-one error usually occurs due to:

- a) Wrong data type
- b) Incorrect loop boundary conditions
- c) Using correct indexes
- d) Proper array size
  **Answer: b**

74. Accessing array elements outside their bounds may cause:

- a) Improved performance
- b) Logical correctness
- c) Runtime errors or crashes
- d) Better memory management
  **Answer: c**

75. Hard-coding values in source code is considered a mistake because it:

- a) Improves flexibility
- b) Simplifies maintenance
- c) Reduces reusability and maintainability
- d) Increases readability
  **Answer: c**

76. Duplicate code blocks in a program usually lead to:

- a) Faster execution
- b) Easier maintenance
- c) Increased chance of errors
- d) Reduced memory usage
  **Answer: c**

77. Poor error handling may cause:

- a) Clear error messages
- b) Graceful system failure
- c) Program crashes

- d) Improved user experience
  **Answer: c**

78. Ignoring return values from functions can result in:

- a) Better abstraction
- b) Loss of important information
- c) Faster execution
- d) Improved reliability
  **Answer: b**

79. Using magic numbers in code refers to:

- a) Using constants with clear names
- b) Using unexplained numeric literals
- c) Using variables instead of numbers
- d) Using configuration files
  **Answer: b**

80. Lack of comments in complex code sections may lead to:

- a) Better performance
- b) Easier understanding
- c) Poor maintainability
- d) Improved readability
  **Answer: c**

81. A common mistake in function design is:

- a) Single responsibility
- b) Clear input/output
- c) Too many responsibilities
- d) Proper parameter naming
  **Answer: c**

82. Tight coupling between modules makes the system:

- a) Easier to modify
- b) More flexible

- c) Harder to maintain and test
- d) More reusable
  **Answer: c**

83. High cohesion in a module means:

- a) The module does many unrelated tasks
- b) The module has closely related responsibilities
- c) The module depends on many other modules
- d) The module is hard to reuse
  **Answer: b**

84. A memory leak occurs when:

- a) Memory is allocated and properly released
- b) Memory is never allocated
- c) Allocated memory is not released after use
- d) Memory usage is optimized
  **Answer: c**

85. Memory leaks typically cause:

- a) Faster execution
- b) Reduced memory usage
- c) Gradual performance degradation
- d) Compilation errors
  **Answer: c**

86. Dangling references occur when:

- a) Memory is still allocated
- b) References point to freed memory
- c) Objects are never created
- d) Variables are global
  **Answer: b**

87. Garbage Collection aims to:

- a) Increase memory fragmentation

- b) Free unused memory automatically
- c) Prevent object creation
- d) Reduce CPU usage only
  **Answer: b**

88. Garbage collection mainly helps in:

- a) Eliminating syntax errors
- b) Managing memory automatically
- c) Improving UI design
- d) Increasing code size
  **Answer: b**

89. Using dynamic memory excessively without control may lead to:

- a) Better performance
- b) Memory fragmentation
- c) Improved reliability
- d) Smaller memory footprint
  **Answer: b**

90. A common mistake related to exception handling is:

- a) Handling all exceptions
- b) Ignoring exceptions
- c) Logging errors properly
- d) Providing meaningful messages
  **Answer: b**

91. Catching generic exceptions instead of specific ones may cause:

- a) Better error diagnosis
- b) Loss of error details
- c) Improved program flow
- d) Faster execution
  **Answer: b**

92. Poor naming conventions in code may result in:

- a) Clear understanding
- b) Improved maintainability
- c) Confusing and unreadable code
- d) Better documentation
  **Answer: c**

93. Debugging usually starts after:

- a) Writing requirements
- b) Executing test cases and detecting defects
- c) Designing architecture
- d) Deploying software
  **Answer: b**

94. The primary goal of debugging is to:

- a) Prevent defects
- b) Find defects
- c) Fix the root cause of defects
- d) Measure software quality
  **Answer: c**

95. Debugging tools are mainly used to:

- a) Write test cases
- b) Monitor program execution
- c) Improve UI design
- d) Measure code size
  **Answer: b**

96. Breakpoints in debugging are used to:

- a) Terminate program execution
- b) Pause execution at specific points
- c) Optimize performance
- d) Allocate memory
  **Answer: b**

97. Step-by-step execution in debugging helps to:

- a) Speed up execution
- b) Observe program behavior closely
- c) Reduce memory usage
- d) Eliminate testing
  **Answer: b**

98. Debugging does NOT aim to:

- a) Identify root causes
- b) Correct errors
- c) Validate requirements
- d) Analyze program behavior
  **Answer: c**

99. Debugging is considered a:

- a) Preventive activity
- b) Corrective activity
- c) Management activity
- d) Documentation activity
  **Answer: b**

100. Testing and debugging are related because:

- a) They are the same activity
- b) Debugging replaces testing
- c) Testing identifies defects and debugging fixes them
- d) Testing fixes defects directly
  **Answer: c**

**MCQ – Lecture 2**

1. Garbage Collection is mainly concerned with:

- a) CPU scheduling
- b) Automatic memory management
- c) Disk optimization
- d) Network bandwidth
  **Answer: b**

2. Garbage Collection aims to:

- a) Increase memory fragmentation
- b) Free unused memory automatically
- c) Prevent object creation
- d) Improve UI performance
  **Answer: b**

3. Garbage Collection is used to:

- a) Allocate memory manually
- b) Remove syntax errors
- c) Reclaim memory occupied by unused objects
- d) Increase execution speed only
  **Answer: c**

4. An object becomes eligible for Garbage Collection when:

- a) It is created
- b) It is referenced by at least one variable
- c) It has no reachable references
- d) The program terminates
  **Answer: c**

5. Garbage Collection helps in preventing:

- a) Compilation errors
- b) Memory leaks
- c) Syntax errors

- d) Logic errors
  **Answer: b**

6. A memory leak occurs when:

- a) Memory is freed too early
- b) Memory is allocated and never released
- c) Objects are garbage collected correctly
- d) Stack memory is used
  **Answer: b**

7. Garbage Collection mainly manages:

- a) Stack memory
- b) CPU registers
- c) Heap memory
- d) Cache memory
  **Answer: c**

8. Heap memory is used to store:

- a) Local variables only
- b) Static variables only
- c) Objects and dynamic data
- d) CPU instructions
  **Answer: c**

9. Stack memory is mainly used for:

- a) Dynamic objects
- b) Function calls and local variables
- c) Long-lived objects
- d) Garbage collection roots
  **Answer: b**

10. Objects stored in heap memory are usually:

- a) Automatically destroyed when scope ends
- b) Managed manually only

- c) Managed by Garbage Collector
- d) Never removed
  **Answer: c**

11. Garbage Collection reduces the need for:

- a) Manual memory deallocation
- b) Testing
- c) Debugging
- d) Code documentation
  **Answer: a**

12. One disadvantage of Garbage Collection is:

- a) Automatic memory management
- b) Program pauses during execution
- c) Reduced code complexity
- d) Fewer memory errors
  **Answer: b**

13. Garbage Collection pauses are also known as:

- a) Runtime errors
- b) Stop-the-world events
- c) Syntax blocks
- d) Deadlocks
  **Answer: b**

14. Stop-the-world means:

- a) Program terminates
- b) Only garbage collector runs while application threads are paused
- c) CPU stops completely
- d) Threads run faster
  **Answer: b**

15. Garbage Collection may affect:

- a) Code readability

- b) Application performance
- c) Software requirements
- d) Source code syntax
  **Answer: b**

16. Memory fragmentation refers to:

- a) Efficient memory usage
- b) Continuous memory allocation
- c) Memory being divided into small unusable pieces
- d) Stack overflow
  **Answer: c**

17. Garbage Collection can help reduce:

- a) Memory fragmentation
- b) CPU utilization
- c) Disk usage
- d) Network latency
  **Answer: a**

18. A reference in programming points to:

- a) A data type
- b) A memory location of an object
- c) A CPU register
- d) A file
  **Answer: b**

19. When an object has no references pointing to it, it is:

- a) Active
- b) Reachable
- c) Eligible for garbage collection
- d) Pinned
  **Answer: c**

20. A dangling reference occurs when:

- a) An object is still referenced
- b) A reference points to freed memory
- c) Memory is properly released
- d) Garbage collector runs
  **Answer: b**

21. Garbage Collection does NOT guarantee:

- a) Automatic memory cleanup
- b) No memory leaks
- c) No performance overhead
- d) Heap memory management
  **Answer: c**

22. Languages that support Garbage Collection include:

- a) C only
- b) C++ only
- c) Java and C#
- d) Assembly only
  **Answer: c**

23. Manual memory management is required in:

- a) Java
- b) C#
- c) C
- d) Python
  **Answer: c**

24. One main benefit of Garbage Collection is:

- a) Faster compilation
- b) Reduced programming errors
- c) Better UI design
- d) Smaller code size
  **Answer: b**

25. Garbage Collection works based on:

- a) Object lifetime
- b) User input
- c) CPU speed
- d) Network usage
  **Answer: a**

26. Root references for Garbage Collection may include:

- a) Local variables
- b) Static variables
- c) Active threads
- d) All of the above
  **Answer: d**

27. An object referenced by a root is considered:

- a) Garbage
- b) Unreachable
- c) Reachable
- d) Invalid
  **Answer: c**

28. Garbage Collection identifies garbage objects by:

- a) Scanning reachable objects
- b) Removing all objects
- c) Checking syntax
- d) Monitoring CPU
  **Answer: a**

29. A common issue when Garbage Collection runs frequently is:

- a) Increased memory availability
- b) Increased CPU overhead
- c) Improved performance

- d) Faster response time
  **Answer: b**

30. Garbage Collection mainly helps developers by:

- a) Eliminating testing
- b) Managing memory automatically
- c) Writing faster code
- d) Avoiding compilation
  **Answer: b**

31. A profiler is mainly used to:

- a) Write source code
- b) Measure program performance during execution
- c) Detect syntax errors
- d) Design databases
  **Answer: b**

32. Profiling is considered a:

- a) Static analysis technique
- b) Dynamic analysis technique
- c) Design technique
- d) Documentation technique
  **Answer: b**

33. Profilers collect information about:

- a) Code formatting
- b) Runtime behavior of applications
- c) Requirement changes
- d) Compilation errors
  **Answer: b**

34. CPU profiling focuses on:

- a) Memory allocation
- b) Disk usage

- c) Processor time consumption
- d) Network traffic
  **Answer: c**

35. Memory profiling is mainly used to:

- a) Detect syntax errors
- b) Analyze memory usage and leaks
- c) Improve UI layout
- d) Measure bandwidth
  **Answer: b**

36. A profiler can help detect:

- a) Infinite loops
- b) Memory leaks
- c) Performance bottlenecks
- d) All of the above
  **Answer: d**

37. A performance bottleneck is:

- a) The fastest part of the system
- b) A resource limiting system performance
- c) A syntax issue
- d) A user interface problem
  **Answer: b**

38. Profilers are typically used during:

- a) Requirements analysis
- b) Design phase only
- c) Program execution
- d) Code documentation
  **Answer: c**

39. Profiling helps developers to:

- a) Guess performance issues

- b) Identify exact locations of performance problems
- c) Eliminate testing
- d) Reduce documentation
  **Answer: b**

40. Time profiling measures:

- a) Memory usage
- b) Disk space
- c) Execution time of methods or functions
- d) Number of users
  **Answer: c**

41. Sampling profilers work by:

- a) Measuring every instruction
- b) Periodically sampling program execution
- c) Analyzing source code only
- d) Monitoring user input
  **Answer: b**

42. Instrumentation-based profiling works by:

- a) Inserting probes into code
- b) Sampling execution randomly
- c) Ignoring runtime behavior
- d) Disabling garbage collection
  **Answer: a**

43. Instrumentation profiling usually has:

- a) No overhead
- b) Very high accuracy with more overhead
- c) Lower accuracy
- d) No performance impact
  **Answer: b**

44. Sampling profilers usually have:

- a) High overhead
- b) Lower overhead
- c) No usefulness
- d) No accuracy
  **Answer: b**

45. Profiling overhead refers to:

- a) Memory freed by GC
- b) Extra cost added by profiling tools
- c) Network delay
- d) Disk latency
  **Answer: b**

46. Profilers can show:

- a) Method call counts
- b) Execution time per method
- c) Memory allocation per object
- d) All of the above
  **Answer: d**

47. A hotspot in profiling means:

- a) A rarely executed code
- b) Code that consumes most resources
- c) Unused functions
- d) Dead code
  **Answer: b**

48. Profiling is typically done:

- a) Before writing code
- b) After performance problems appear
- c) Instead of testing
- d) Only after deployment
  **Answer: b**

49. One risk of premature optimization is:

- a) Improved maintainability
- b) Wasted effort on non-critical code
- c) Better system performance
- d) Reduced complexity
  **Answer: b**

50. The correct performance optimization process is:

- a) Optimize first, then measure
- b) Measure, identify bottleneck, then optimize
- c) Optimize everything
- d) Skip measurement
  **Answer: b**

51. Profiling helps avoid:

- a) Guess-based optimization
- b) Measuring performance
- c) Debugging
- d) Testing
  **Answer: a**

52. Memory leaks identified by profilers usually occur in:

- a) Stack memory
- b) Heap memory
- c) CPU cache
- d) Registers
  **Answer: b**

53. Profilers can visualize data using:

- a) Flowcharts
- b) Graphs and charts
- c) UML diagrams

- d) ER diagrams
  **Answer: b**

54. Profiling tools are commonly used with:

- a) Static analysis only
- b) Runtime monitoring
- c) Design reviews
- d) Requirement validation
  **Answer: b**

55. Profiling results should be interpreted:

- a) Without context
- b) Along with workload and environment
- c) Only by managers
- d) Ignoring system load
  **Answer: b**

56. Garbage Collection and Profiling are related because:

- a) Profilers control garbage collection
- b) Profilers help analyze GC behavior and memory usage
- c) GC replaces profiling
- d) Profiling prevents GC
  **Answer: b**

57. Profiling is most useful when:

- a) System requirements are unclear
- b) Performance issues are reported
- c) Code is not written yet
- d) Documentation is missing
  **Answer: b**

58. Excessive profiling may cause:

- a) Faster execution
- b) Reduced performance accuracy

- c) High overhead affecting measurements
- d) Elimination of bugs
  **Answer: c**

59. A well-used profiler helps achieve:

- a) Random optimization
- b) Targeted performance improvement
- c) Increased code size
- d) Reduced testing
  **Answer: b**

60. Performance tuning should focus on:

- a) All parts equally
- b) Identified bottlenecks only
- c) UI components
- d) Rarely used code
  **Answer: b**

**MCQ – Lecture 3-4**

1. Software Testing is defined as:
   a) Static verification of code correctness without execution
   b) Dynamic verification of program behavior using selected test cases against expected behavior
   c) Manual inspection of documentation only
   d) Compilation and deployment of software
   **Answer: b**

2. The term *Dynamic* in software testing means:
   a) Tests are written dynamically at runtime
   b) Testing depends on changing requirements
   c) The program is executed using actual input values
   d) Tests are generated automatically
   **Answer: c**

3. The term *Finite* in software testing refers to:
   a) Test cases are limited by hardware
   b) Only a limited number of test cases can be executed
   c) Input domain is always small
   d) Testing stops after deployment
   **Answer: b**

4. The execution domain of a program is usually:
   a) Finite
   b) Empty
   c) Infinite
   d) Equal to number of test cases
   **Answer: c**

5. The word *Selected* in the testing definition refers to:
   a) Random selection only
   b) Tester experience only
   c) Test techniques used to choose test cases
   d) Compiler optimization
   **Answer: c**

6. Different test case selection criteria may lead to:
   a) Same effectiveness
   b) Same number of defects
   c) Vastly different effectiveness
   d) No impact on results
   **Answer: c**

7. The term *Expected* behavior refers to:
   a) Actual program output
   b) Developer assumptions
   c) Decision process to judge acceptability of outputs
   d) Performance metrics
   **Answer: c**

8. Black Box testing is also known as:
   a) Structural testing
   b) Glass box testing
   c) Functional testing
   d) Gray box testing
   **Answer: c**

9. Black Box testing focuses mainly on:
   a) Source code structure
   b) Internal logic
   c) Outputs generated from inputs
   d) Control flow graphs
   **Answer: c**

10. In Black Box testing, the tester has:
    a) Full understanding of code
    b) Partial understanding of code
    c) No insight into code and architecture
    d) Access to source code only
    **Answer: c**

11. White Box testing is also called:
    a) Functional testing
    b) Behavioral testing
    c) Structural testing
    d) Acceptance testing
    **Answer: c**

12. White Box testing takes into account:
    a) Only requirements
    b) Only outputs
    c) Full internal code and architecture
    d) User experience
    **Answer: c**

13. Gray Box testing provides:
    a) No interaction with code

b) Full interaction with code
c) Partial interaction with internal code
d) Only UI testing
**Answer: c**

14. Static Analysis Techniques are based on:
    a) Program execution
    b) Runtime monitoring
    c) Examination of documents and code without execution
    d) User feedback
    **Answer: c**

15. Static analysis can be applied during:
    a) Implementation phase only
    b) Testing phase only
    c) Requirements, design, and implementation phases
    d) Maintenance phase only
    **Answer: c**

16. Traditional static techniques are generally:
    a) Fully automated
    b) Heavily manual and time consuming
    c) Fast and precise
    d) Dynamic
    **Answer: b**

17. Software inspection is defined as:
    a) Automated execution of test cases
    b) Step-by-step document analysis using defect checklists
    c) Performance testing
    d) Stress testing
    **Answer: b**

18. Software reviews involve:
    a) Only developers
    b) Only testers
    c) Presenting work products to stakeholders for feedback
    d) Compiler validation
    **Answer: c**

19. Code reading aims to discover:
    a) Runtime exceptions
    b) Performance bottlenecks
    c) Typing errors that do not violate syntax

d) Memory leaks
**Answer: c**

20. Algorithm analysis focuses on:
a) UI behavior
b) Code style
c) Complexity and worst/average case analysis
d) Test automation
**Answer: c**

21. Dynamic testing techniques are based on:
a) Documentation review
b) Program execution
c) Code comments
d) UML diagrams
**Answer: b**

22. Profiling records:
a) Compilation errors
b) UI layout issues
c) Frequency of program events during execution
d) User feedback
**Answer: c**

23. Unit testing targets:
a) Entire system
b) Interfaces between modules
c) Smallest testable piece of software
d) User workflows
**Answer: c**

24. A unit generally represents:
a) Work of multiple teams
b) Result of one programmer's work
c) Entire subsystem
d) Full application
**Answer: b**

25. Integration testing focuses on:
a) Individual modules
b) User acceptance
c) Combined modules interaction
d) UI design
**Answer: c**

26. Integration testing is usually performed:
    a) Before unit testing
    b) After unit testing
    c) After acceptance testing
    d) Before coding
    **Answer: b**

27. Unit testing is typically a:
    a) Black box technique
    b) Gray box technique
    c) White box technique
    d) Non-functional technique
    **Answer: c**

28. Integration testing is typically a:
    a) White box technique
    b) Black box technique
    c) Static technique
    d) Manual-only technique
    **Answer: b**

29. System testing involves:
    a) Individual functions
    b) Entire system in actual hardware environment
    c) Only database layer
    d) Only UI
    **Answer: b**

30. Acceptance testing mainly focuses on:
    a) Code coverage
    b) Performance tuning
    c) Usability and user acceptance
    d) Memory optimization
    **Answer: c**

31. Validation testing ensures that:
    a) Code follows style rules
    b) Algorithms are optimal
    c) Software meets customer requirements
    d) Tests are automated
    **Answer: c**

32. Random testing selects inputs:
    a) Based on boundaries only

b) Using fixed equivalence classes
c) Purely randomly with probabilities
d) From invalid partitions only
**Answer: c**

33. Partition testing divides input domain into:
a) Random sets
b) Sub-domains with equivalent behavior
c) Execution paths
d) Code blocks
**Answer: b**

34. Structural testing is also known as:
a) Specification-based testing
b) Code-based testing
c) Acceptance testing
d) Regression testing
**Answer: b**

35. Equivalence Partitioning reduces test cases because:
a) All values behave differently
b) One value represents the entire partition
c) Boundaries are ignored
d) Only invalid inputs are tested
**Answer: b**

36. Boundary Value Analysis focuses on testing:
a) Middle values only
b) Random values
c) Values at edges of partitions
d) Performance limits
**Answer: c**

37. If valid month range is 1–12, value 0 belongs to:
a) Valid partition
b) Invalid partition
c) Boundary value
d) Middle value
**Answer: b**

38. Test-first programming means:
a) Writing code before tests
b) Writing tests after deployment
c) Writing tests before coding

d) Skipping design
**Answer: c**

39. The main goal of Test-First Programming is to:
a) Increase documentation
b) Make development lighter and simpler
c) Remove testing phase
d) Replace developers
**Answer: b**

40. Verification answers the question:
a) Does it meet user needs?
b) Is it fast enough?
c) Are algorithms implemented correctly?
d) Is UI user-friendly?
**Answer: c**

41. Validation answers the question:
a) Is code optimized?
b) Does it meet customer requirements?
c) Is syntax correct?
d) Is memory usage low?
**Answer: b**

42. A test case consists of:
a) Code only
b) Input – processing – output
c) UI – database – server
d) Bug report only
**Answer: b**

43. If Actual Results equal Expected Results, the test case is:
a) Blocked
b) Skipped
c) Failed
d) Passed
**Answer: d**

44. In Top-Down testing, missing lower modules are replaced by:
a) Drivers
b) Compilers
c) Stubs
d) Mocks only
**Answer: c**

45. Bottom-Up testing requires:
    a) Stubs only
    b) Drivers only
    c) Both stubs and drivers
    d) No dummy modules
    **Answer: b**

46. Smoke testing is best described as:
    a) Full regression testing
    b) Performance testing
    c) Rapid test of major functionality
    d) Security testing
    **Answer: c**

47. Alpha testing is conducted:
    a) At user environment without developers
    b) At developer environment with end users
    c) After product release
    d) By testers only
    **Answer: b**

48. Beta testing is conducted:
    a) In controlled environment
    b) Before coding
    c) At end-user environment
    d) Only by developers
    **Answer: c**

49. In TDD, RED phase means:
    a) Refactor code
    b) Make test pass
    c) Create failing test
    d) Deploy software
    **Answer: c**

50. In TDD, GREEN phase means:
    a) Improve design
    b) Make the test pass
    c) Remove tests
    d) Write documentation
    **Answer: b**

51. In TDD, the REFACTOR phase aims to:
    a) Add new features

b) Improve code structure without changing behavior
c) Remove test cases
d) Delay testing
**Answer: b**

52. A major benefit of TDD is:
a) Writing less tests
b) Early detection of defects
c) Eliminating documentation
d) Skipping integration testing
**Answer: b**
53. Black-box testing derives test cases from:
a) Source code
b) Control flow graph
c) Requirements and specifications
d) Compiler errors
**Answer: c**
54. White-box testing derives test cases from:
a) User manuals
b) Requirements only
c) Internal code structure
d) UI screens
**Answer: c**
55. Gray-box testing combines knowledge from:
a) Requirements only
b) Code only
c) Both requirements and internal structure
d) Performance metrics only
**Answer: c**
56. Static testing techniques include:
a) Unit testing
b) System testing
c) Code inspection
d) Load testing
**Answer: c**
57. A walkthrough is best described as:
a) Formal review with checklist
b) Informal review led by the author
c) Automated analysis
d) Performance measurement
**Answer: b**
58. An inspection differs from a walkthrough because inspection is:
a) Informal
b) Automated
c) Formal and checklist-based
d) Performed by users
**Answer: c**

59. One main objective of reviews is to:
    a) Fix defects
    b) Identify defects early
    c) Execute test cases
    d) Measure performance
    **Answer: b**
60. Static testing helps in:
    a) Detecting runtime failures
    b) Finding defects before execution
    c) Measuring response time
    d) Stressing the system
    **Answer: b**
61. Dynamic testing can detect:
    a) Syntax errors only
    b) Runtime errors and failures
    c) Documentation defects only
    d) Design inconsistencies only
    **Answer: b**
62. Unit testing is most effective when tests are:
    a) Written after system testing
    b) Written by end users
    c) Automated
    d) Ignored during refactoring
    **Answer: c**
63. A test driver is used in:
    a) Top-down integration testing
    b) Bottom-up integration testing
    c) Acceptance testing
    d) Regression testing
    **Answer: b**
64. A stub is used in:
    a) Bottom-up testing
    b) System testing
    c) Top-down integration testing
    d) Performance testing
    **Answer: c**
65. The main role of a stub is to:
    a) Call lower-level modules
    b) Simulate called modules
    c) Execute real database calls
    d) Replace test cases
    **Answer: b**
66. The main role of a driver is to:
    a) Simulate higher-level modules
    b) Replace UI
    c) Measure performance

d) Replace database
**Answer: a**
67. Regression testing is usually performed when:
a) New requirements are written
b) Code changes are introduced
c) System is deployed
d) Documentation is updated
**Answer: b**
68. Regression testing helps to ensure that:
a) New bugs are introduced
b) Old functionality still works
c) Performance is optimal
d) UI is redesigned
**Answer: b**
69. System testing is usually a:
a) White-box testing
b) Black-box testing
c) Static testing
d) Informal review
**Answer: b**
70. Acceptance testing is mainly concerned with:
a) Code coverage
b) User satisfaction
c) Internal architecture
d) Compiler optimization
**Answer: b**
71. A test oracle is used to:
a) Generate test cases
b) Execute test scripts
c) Decide whether output is correct
d) Measure performance
**Answer: c**
72. Manual testing is characterized by:
a) Automation tools
b) Human execution of test cases
c) No documentation
d) No defects
**Answer: b**
73. Automated testing is best suited for:
a) One-time tests
b) Exploratory testing only
c) Repetitive regression tests
d) Usability testing only
**Answer: c**
74. One limitation of automated testing is:
a) High speed
b) Repeatability

c) High initial setup cost
d) Accuracy
**Answer: c**

75. Test coverage refers to:
a) Number of testers
b) Percentage of code or requirements tested
c) Number of test cases
d) Execution speed
**Answer: b**
76. Statement coverage measures:
a) Execution of all paths
b) Execution of all statements
c) Execution of all branches
d) Execution of all conditions
**Answer: b**
77. Branch coverage ensures that:
a) Every statement is executed
b) Every decision outcome is executed
c) Every loop runs once
d) Every class is instantiated
**Answer: b**
78. Path coverage is generally:
a) Easier than statement coverage
b) Easier than branch coverage
c) Harder due to number of paths
d) Independent of code complexity
**Answer: c**
79. High test coverage guarantees:
a) No defects
b) Correct requirements
c) Better confidence, not absence of bugs
d) Perfect software
**Answer: c**
80. A test plan mainly describes:
a) Code structure
b) Testing scope, approach, and schedule
c) Debugging steps
d) User stories
**Answer: b**
81. Test cases are derived mainly from:
a) Compiler output
b) Requirements and design documents
c) Hardware specs
d) Database schema only
**Answer: b**
82. Exploratory testing is characterized by:
a) Predefined scripts only

b) Simultaneous learning and testing
c) No tester involvement
d) Automation only
**Answer: b**

83. One advantage of exploratory testing is:
a) Complete coverage
b) Flexibility and creativity
c) Repeatability
d) Automation
**Answer: b**

84. One disadvantage of exploratory testing is:
a) Finding critical bugs
b) Low tester skill requirement
c) Difficulty in reproducibility
d) High documentation
**Answer: c**

85. Test environment should ideally be:
a) Completely different from production
b) Similar to production environment
c) Only local machine
d) Ignored
**Answer: b**

86. NUnit is considered a:
a) Performance testing tool
b) Unit testing framework
c) Integration testing framework
d) Debugging tool
**Answer: b**

87. NUnit is mainly used with:
a) Java
b) Python
c) .NET languages
d) C only
**Answer: c**

88. NUnit supports testing at the:
a) System level only
b) Acceptance level only
c) Unit level
d) Network level
**Answer: c**

89. A test fixture in NUnit represents:
a) A single test method
b) A class that contains test methods
c) A configuration file
d) A test result
**Answer: b**

90. The attribute used to mark a test class in NUnit is:
    a) [Test]
    b) [TestFixture]
    c) [SetUp]
    d) [Ignore]
    **Answer: b**
91. The [Test] attribute is used to:
    a) Mark a class
    b) Mark a method as a test case
    c) Initialize objects
    d) Clean resources
    **Answer: b**
92. A method marked with [Test] must be:
    a) Private
    b) Static
    c) Public
    d) Protected
    **Answer: c**
93. The [SetUp] method in NUnit is executed:
    a) Once before all tests
    b) Once after all tests
    c) Before each test method
    d) After each test method
    **Answer: c**
94. The [TearDown] method in NUnit is executed:
    a) Before each test
    b) After each test
    c) Before all tests
    d) Only if test fails
    **Answer: b**
95. The attribute used to run code once before all tests is:
    a) [SetUp]
    b) [OneTimeSetUp]
    c) [Test]
    d) [Ignore]
    **Answer: b**
96. The attribute used to run code once after all tests is:
    a) [TearDown]
    b) [OneTimeTearDown]
    c) [TestFixture]
    d) [Category]
    **Answer: b**
97. Assertions in NUnit are used to:
    a) Execute test cases
    b) Compare expected and actual results
    c) Initialize objects

d) Measure performance
**Answer: b**

98. Assert.AreEqual(expected, actual) checks:
a) Reference equality
b) Value equality
c) Memory address
d) Data type only
**Answer: b**

99. Assert.IsTrue(condition) will pass when:
a) Condition is false
b) Condition throws exception
c) Condition evaluates to true
d) Condition is null
**Answer: c**

100. Assert.IsFalse(condition) will pass when:
a) Condition is true
b) Condition evaluates to false
c) Condition is null
d) Condition throws exception
**Answer: b**

101. Assert.IsNull(object) passes when:
a) Object is initialized
b) Object equals zero
c) Object is null
d) Object is empty
**Answer: c**

102. Assert.IsNotNull(object) passes when:
a) Object is null
b) Object has value
c) Object is false
d) Object is empty
**Answer: b**

103. If an assertion fails, NUnit will:
a) Skip remaining tests
b) Stop the entire test suite
c) Mark the test as failed
d) Ignore the error
**Answer: c**

104. Multiple assertions in one test method:
a) Are not allowed
b) Stop execution after first failure
c) Are executed sequentially
d) Are ignored
**Answer: c**

105. NUnit test results can be:
a) Passed only
b) Failed only

c) Passed, Failed, Ignored
d) Compiled
**Answer: c**

106. The [Ignore] attribute is used to:
a) Delete a test
b) Temporarily skip a test
c) Mark test as failed
d) Run test later
**Answer: b**

107. NUnit test execution is usually done by:
a) Compiler
b) NUnit Test Runner
c) End user
d) Operating system
**Answer: b**

108. Automated unit tests are especially useful for:
a) UI design
b) Regression testing
c) Requirement gathering
d) Deployment
**Answer: b**

109. NUnit helps improve code quality by:
a) Eliminating need for testing
b) Encouraging frequent testing
c) Replacing debugging
d) Writing documentation
**Answer: b**

110. Unit tests written using NUnit should be:
a) Dependent on each other
b) Independent and isolated
c) Executed manually only
d) Written after deployment
**Answer: b**

111. A good unit test should:
a) Test multiple modules together
b) Depend on external systems
c) Test one behavior
d) Be very long
**Answer: c**

112. Mock objects are used to:
a) Replace test cases
b) Simulate real dependencies
c) Compile code
d) Measure performance
**Answer: b**

113. Using mocks helps to:
a) Increase coupling

b) Isolate the unit under test
c) Reduce automation
d) Replace assertions
**Answer: b**

114. NUnit supports:
a) Manual testing only
b) Automated testing
c) Performance profiling
d) Security testing
**Answer: b**

115. Unit testing with NUnit is usually done:
a) After system testing
b) During coding
c) After deployment
d) By end users
**Answer: b**

**MCQ – Lecture 5**

1. Software quality is best defined as:

- a) Number of features in the system
- b) Degree to which software meets stated and implied requirements
- c) Execution speed only
- d) Absence of documentation
  **Answer: b**

2. Software quality metrics are mainly used to:

- a) Write code
- b) Measure and evaluate software quality attributes
- c) Replace testing
- d) Improve hardware performance
  **Answer: b**

3. A metric is:

- a) A testing tool
- b) A quantitative measure of an attribute
- c) A debugging technique
- d) A design pattern
  **Answer: b**

4. Software metrics help managers to:

- a) Guess project status
- b) Make informed decisions
- c) Avoid planning
- d) Eliminate testing
  **Answer: b**

5. Product metrics measure:

- a) Development process only
- b) Characteristics of the software product
- c) Team productivity
- d) Project schedule
  **Answer: b**

6. Process metrics focus on:

- a) Software size
- b) How the software is developed
- c) User satisfaction

- d) Memory usage
  **Answer: b**

7. Project metrics are used to:

- a) Measure code quality
- b) Track project progress and productivity
- c) Detect runtime errors
- d) Execute test cases
  **Answer: b**

8. Direct software measures include:

- a) Maintainability
- b) Usability
- c) Lines of Code (LOC)
- d) Quality
  **Answer: c**

9. Indirect software measures include:

- a) LOC
- b) Function points
- c) Quality and maintainability
- d) Execution time
  **Answer: c**

10. Lines of Code (LOC) is considered:

- a) Language independent
- b) A direct size-oriented metric
- c) A usability metric
- d) A performance metric
  **Answer: b**

11. A disadvantage of LOC is that it is:

- a) Easy to calculate
- b) Language dependent
- c) Widely used
- d) Simple to understand
  **Answer: b**

12. Size-oriented metrics normalize data using:

- a) Number of users
- b) LOC
- c) Execution time

- d) Cost
  **Answer: b**

13. Errors per KLOC is an example of:

- a) Function-oriented metric
- b) Size-oriented metric
- c) Performance metric
- d) Process metric
  **Answer: b**

14. Function Point metrics are:

- a) Based on programming language
- b) Language independent
- c) Based on LOC only
- d) Hardware dependent
  **Answer: b**

15. Function Points are derived from:

- a) Code structure
- b) Information domain characteristics
- c) CPU usage
- d) Memory allocation
  **Answer: b**

16. Function-oriented metrics are useful early because they:

- a) Require complete code
- b) Can be estimated from requirements
- c) Depend on compiler
- d) Ignore functionality
  **Answer: b**

17. Object-oriented metrics may include:

- a) Number of classes
- b) Number of scenarios
- c) Coupling between objects
- d) All of the above
  **Answer: d**

18. Use-case oriented metrics are based on:

- a) LOC
- b) Execution paths
- c) Use cases and scenarios

- d) CPU cycles
  **Answer: c**

19. A problem with use-case metrics is:

- a) Language dependency
- b) No standard size for use cases
- c) High accuracy
- d) Ease of calculation
  **Answer: b**

20. Software quality attributes include:

- a) Correctness
- b) Reliability
- c) Maintainability
- d) All of the above
  **Answer: d**

21. Correctness is usually measured by:

- a) Execution time
- b) Defects per KLOC
- c) Memory usage
- d) Throughput
  **Answer: b**

22. Reliability is related to:

- a) Number of failures over time
- b) UI design
- c) Code readability
- d) Documentation size
  **Answer: a**

23. Mean Time To Failure (MTTF) measures:

- a) Time to fix a bug
- b) Average time between failures
- c) CPU idle time
- d) Development time
  **Answer: b**

24. Maintainability measures how easily software can be:

- a) Executed
- b) Modified and corrected
- c) Compiled

- d) Installed
  **Answer: b**

25. Mean Time To Change (MTTC) is used to measure:

- a) Performance
- b) Maintainability
- c) Reliability
- d) Usability
  **Answer: b**

26. Defect Removal Efficiency (DRE) is calculated as:

- a) D / E
- b) E / (E + D)
- c) D / (E + D)
- d) E × D
  **Answer: b**

27. A high DRE value indicates:

- a) Poor testing
- b) Many escaped defects
- c) Effective defect removal
- d) Low quality
  **Answer: c**

28. Usability refers to:

- a) Code complexity
- b) Ease of use for end users
- c) Execution speed
- d) Security strength
  **Answer: b**

29. Web applications are different from traditional apps because they:

- a) Run on single machine
- b) Use client-server architecture
- c) Do not need testing
- d) Are static
  **Answer: b**

30. Web testing focuses on:

- a) Hardware drivers
- b) Browsers, servers, and networks
- c) Desktop UI only

- d) Compiler behavior
  **Answer: b**

31. Compatibility testing in web applications ensures:

- a) Same code size
- b) Proper behavior across browsers and platforms
- c) Faster execution
- d) No server load
  **Answer: b**

32. Web usability testing focuses on:

- a) Database queries
- b) User experience and navigation
- c) Memory leaks
- d) CPU utilization
  **Answer: b**

33. Web performance testing measures:

- a) Code style
- b) Response time and throughput
- c) UI colors
- d) Documentation quality
  **Answer: b**

34. Load testing checks system behavior under:

- a) Normal load
- b) Heavy expected load
- c) Failure conditions
- d) Zero users
  **Answer: b**

35. Stress testing checks system behavior:

- a) Under normal conditions
- b) Beyond maximum capacity
- c) Without users
- d) During development only
  **Answer: b**

36. Response time is:

- a) Time taken by server to process request only
- b) Time from request submission to response received
- c) Network latency only

- d) CPU processing time
  **Answer: b**

37. Latency refers to:

- a) Server processing time
- b) Network delay
- c) Rendering time
- d) Database execution time
  **Answer: b**

38. Service Time can be calculated as:

- a) Response Time + Latency
- b) Response Time – Latency
- c) Latency – Response Time
- d) Throughput × Time
  **Answer: b**

39. Throughput measures:

- a) Number of requests processed per unit time
- b) Execution time of one request
- c) Memory usage
- d) Code complexity
  **Answer: a**

40. Bandwidth refers to:

- a) CPU speed
- b) Bits per second transferred
- c) Memory size
- d) Disk space
  **Answer: b**

41. Capacity-based utilization is:

- a) Harder to calculate than time-based
- b) Easier to calculate than time-based
- c) Same as latency
- d) Independent of load
  **Answer: b**

42. Time-based utilization measures:

- a) Percentage of busy time
- b) Disk usage
- c) Network speed

- d) Number of users
**Answer: a**

43. Web testing must consider:

- a) Client side only
- b) Server side only
- c) Client, server, and network
- d) Database only
**Answer: c**

44. Security testing in web applications aims to:

- a) Improve UI
- b) Prevent unauthorized access
- c) Increase speed
- d) Reduce code size
**Answer: b**

45. SQL Injection is a:

- a) Performance issue
- b) Security vulnerability
- c) UI bug
- d) Hardware fault
**Answer: b**

46. Cross-site scripting (XSS) targets:

- a) Server memory
- b) Client-side scripts
- c) Database storage
- d) Network bandwidth
**Answer: b**

47. Web testing differs from desktop testing due to:

- a) Single user
- b) Multi-tier architecture
- c) Lack of browsers
- d) No network
**Answer: b**

48. A major challenge in web testing is:

- a) Static content
- b) Browser diversity
- c) Single platform

- d) No users
  **Answer: b**

49. Session management testing focuses on:

- a) UI layout
- b) User authentication and sessions
- c) Code compilation
- d) Database indexing
  **Answer: b**

50. Quality metrics in web testing help to:

- a) Eliminate bugs completely
- b) Quantify and evaluate quality
- c) Replace manual testing
- d) Skip documentation
  **Answer: b**

MCQ – Lecture 6

1. Web security is mainly concerned with:

- a) Improving UI design
- b) Protecting web applications from attacks
- c) Increasing response time
- d) Reducing code size
  **Answer: b**

2. A web security vulnerability is:

- a) A performance bottleneck
- b) A weakness that can be exploited by attackers
- c) A UI issue
- d) A documentation error
  **Answer: b**

3. Security testing aims to:

- a) Improve usability
- b) Detect and prevent security vulnerabilities
- c) Increase execution speed
- d) Reduce memory usage
  **Answer: b**

4. Authentication is the process of:

- a) Granting permissions
- b) Verifying user identity
- c) Encrypting data
- d) Logging user actions
  **Answer: b**

5. Authorization is the process of:

- a) Verifying user credentials
- b) Identifying a user
- c) Granting access rights

- d) Encrypting passwords
  **Answer: c**

6. A strong authentication mechanism should:

- a) Use plain text passwords
- b) Rely on user names only
- c) Use secure credentials and validation
- d) Avoid encryption
  **Answer: c**

7. Session management is used to:

- a) Store UI settings
- b) Maintain user state across requests
- c) Compile server code
- d) Improve page layout
  **Answer: b**

8. HTTP is considered:

- a) Stateful
- b) Stateless
- c) Persistent
- d) Secure by default
  **Answer: b**

9. Because HTTP is stateless, web applications use:

- a) Threads
- b) Sessions
- c) Compilers
- d) Cookies only for styling
  **Answer: b**

10. A session ID is:

- a) A database key
- b) A unique identifier for a user session

- c) A user password
- d) An IP address
  **Answer: b**

11. A web session ID should be:

- a) Predictable
- b) Short and simple
- c) Unique and hard to guess
- d) Based on user name
  **Answer: c**

12. Cookies are mainly used to:

- a) Execute scripts
- b) Store session-related data on client side
- c) Encrypt database
- d) Improve performance
  **Answer: b**

13. Cookies are stored:

- a) On the server only
- b) In database only
- c) On the client side (browser)
- d) In application memory only
  **Answer: c**

14. A security risk of cookies is that they:

- a) Are always encrypted
- b) Can be stolen or manipulated
- c) Cannot store data
- d) Expire immediately
  **Answer: b**

15. Hidden form fields are:

- a) Client-side session management techniques

- b) Server-side session management techniques
- c) Database storage methods
- d) Encryption techniques
  **Answer: a**

16. URL rewriting is:

- a) Encrypting URLs
- b) Passing session ID through URL parameters
- c) Changing page layout
- d) Compressing URLs
  **Answer: b**

17. A disadvantage of URL rewriting is:

- a) Difficult implementation
- b) Session ID exposure in browser history
- c) High performance cost
- d) Incompatibility with browsers
  **Answer: b**

18. Server-side session management stores session data:

- a) On client browser
- b) On server
- c) In URL
- d) In HTML pages
  **Answer: b**

19. A benefit of server-side session management is:

- a) Higher security
- b) Lower server load
- c) No memory usage
- d) Faster client performance
  **Answer: a**

20. SQL Injection is an attack where:

- a) SQL queries are optimized
- b) Attacker injects SQL commands through input fields
- c) Database is backed up
- d) Queries are encrypted
  **Answer: b**

21. SQL Injection mainly targets:

- a) Web server memory
- b) Database
- c) Client browser
- d) Network bandwidth
  **Answer: b**

22. One cause of SQL Injection is:

- a) Using parameterized queries
- b) Input validation
- c) Lack of input validation
- d) Using ORM frameworks
  **Answer: c**

23. A common SQL Injection prevention technique is:

- a) Using plain SQL strings
- b) Using parameterized queries
- c) Storing passwords in plain text
- d) Disabling database
  **Answer: b**

24. Stored procedures:

- a) Increase SQL injection risk
- b) Help prevent SQL injection if used correctly
- c) Are client-side scripts
- d) Replace input validation
  **Answer: b**

25. Cross-Site Scripting (XSS) occurs when:

- a) Server executes malicious SQL
- b) Malicious scripts are injected into web pages
- c) Database is corrupted
- d) Network traffic is intercepted
  **Answer: b**

26. XSS mainly affects:

- a) Server-side logic
- b) Client-side browsers
- c) Database schema
- d) Network routers
  **Answer: b**

27. Reflected XSS occurs when:

- a) Malicious script is stored permanently
- b) Script is reflected immediately in response
- c) Script is executed on server
- d) Script is encrypted
  **Answer: b**

28. Stored XSS occurs when:

- a) Script is stored in database and served to users
- b) Script is executed once
- c) Script runs on server only
- d) Script is blocked by browser
  **Answer: a**

29. Input validation helps prevent:

- a) UI bugs
- b) Security attacks
- c) Performance issues

- d) Compilation errors
  **Answer: b**

30. Output encoding is used to:

- a) Improve UI
- b) Prevent XSS attacks
- c) Speed up server
- d) Encrypt database
  **Answer: b**

31. Cross-Site Request Forgery (CSRF) is an attack where:

- a) User sends valid request intentionally
- b) Attacker tricks user into sending unwanted requests
- c) Database is accessed directly
- d) Server crashes
  **Answer: b**

32. CSRF exploits the fact that:

- a) HTTP is encrypted
- b) Browser automatically sends authentication credentials
- c) Users know SQL
- d) Cookies are disabled
  **Answer: b**

33. A common CSRF prevention technique is:

- a) Using hidden SQL queries
- b) Using CSRF tokens
- c) Disabling cookies
- d) Increasing bandwidth
  **Answer: b**

34. HTTPS provides:

- a) Faster response time
- b) Encrypted communication

- c) UI optimization
- d) Better testing coverage
  **Answer: b**

35. HTTPS is based on:

- a) FTP
- b) SSL/TLS
- c) UDP
- d) SMTP
  **Answer: b**

36. Encryption helps to ensure:

- a) Availability only
- b) Confidentiality of data
- c) UI consistency
- d) Faster execution
  **Answer: b**

37. Passwords should be stored:

- a) In plain text
- b) Encrypted or hashed
- c) In cookies
- d) In URLs
  **Answer: b**

38. Hashing differs from encryption because hashing is:

- a) Reversible
- b) One-way
- c) Faster to decrypt
- d) Used for UI
  **Answer: b**

39. Brute force attack attempts to:

- a) Inject scripts

- b) Guess credentials repeatedly
- c) Break encryption mathematically
- d) Modify database schema
  **Answer: b**

40. Account lockout mechanisms help prevent:

- a) XSS
- b) SQL Injection
- c) Brute force attacks
- d) CSRF
  **Answer: c**

41. Security testing should be performed:

- a) Only after deployment
- b) Only during design
- c) Throughout the SDLC
- d) After system testing only
  **Answer: c**

42. Penetration testing is:

- a) Functional testing
- b) Simulated attack on system
- c) Performance testing
- d) Unit testing
  **Answer: b**

43. OWASP is:

- a) A programming language
- b) A web framework
- c) An organization focused on web security
- d) A database system
  **Answer: c**

44. OWASP Top 10 represents:

- a) Fastest websites
- b) Most common web vulnerabilities
- c) Best testing tools
- d) UI design rules
  **Answer: b**

45. Input sanitization refers to:

- a) Removing unnecessary UI elements
- b) Cleaning and validating user input
- c) Encrypting database
- d) Logging user actions
  **Answer: b**

46. A secure web application should:

- a) Trust user input
- b) Validate all user input
- c) Disable error handling
- d) Avoid testing
  **Answer: b**

47. Error messages in production should:

- a) Show full stack trace
- b) Reveal database details
- c) Be generic and non-informative
- d) Be ignored
  **Answer: c**

48. Logging security events helps in:

- a) UI design
- b) Incident detection and analysis
- c) Reducing memory
- d) Increasing speed
  **Answer: b**

49. Security misconfiguration may result from:

- a) Secure defaults
- b) Unnecessary open services
- c) Strong passwords
- d) Input validation
  **Answer: b**

50. Web security testing ensures that:

- a) Application is bug-free
- b) Application resists common attacks
- c) Performance is optimal
- d) UI is attractive
  **Answer: b**

**MCQ – Lecture 7**

1. Performance testing is mainly used to:

- a) Detect syntax errors
- b) Evaluate system behavior under load
- c) Improve UI design
- d) Validate requirements
  **Answer: b**

2. Performance testing focuses on:

- a) Functional correctness
- b) Speed, scalability, and stability
- c) Code style
- d) Security vulnerabilities
  **Answer: b**

3. A performance issue is usually related to:

- a) Compilation errors
- b) Resource usage and response time
- c) UI colors
- d) Missing requirements
  **Answer: b**

4. Load testing checks system behavior:

- a) With zero users
- b) Under expected normal load
- c) Beyond maximum capacity
- d) Without database
  **Answer: b**

5. Stress testing checks system behavior:

- a) Under normal conditions
- b) With minimum load

- c) Beyond system capacity
- d) With one user only
  **Answer: c**

6. Spike testing is concerned with:

- a) Gradual load increase
- b) Sudden increase or decrease in load
- c) Long duration testing
- d) UI responsiveness
  **Answer: b**

7. Endurance (Soak) testing focuses on:

- a) Short-term load
- b) Long-term system stability
- c) UI layout
- d) Compilation time
  **Answer: b**

8. Volume testing evaluates:

- a) Number of users
- b) Large amount of data handling
- c) CPU speed
- d) Network protocols
  **Answer: b**

9. Scalability testing determines:

- a) Maximum code size
- b) Ability to handle increased load
- c) UI flexibility
- d) Compiler efficiency
  **Answer: b**

10. Performance testing is considered a:

- a) Functional testing

- b) Non-functional testing
- c) Unit testing
- d) Static testing
  **Answer: b**

11. Response time is defined as:

- a) Time for CPU execution only
- b) Time between request and response
- c) Network latency only
- d) Database execution time only
  **Answer: b**

12. Throughput measures:

- a) Time per request
- b) Number of requests processed per unit time
- c) Memory consumption
- d) CPU utilization
  **Answer: b**

13. Latency refers to:

- a) Processing time on server
- b) Network delay
- c) Client rendering time
- d) Disk I/O time
  **Answer: b**

14. Bandwidth represents:

- a) Number of users
- b) Bits transferred per second
- c) CPU frequency
- d) Memory size
  **Answer: b**

15. Service time is calculated as:

- a) Response time + latency
- b) Response time − latency
- c) Throughput × time
- d) Latency − response time
  **Answer: b**

16. Capacity-based utilization is:

- a) Hard to calculate
- b) Easier than time-based utilization
- c) Same as latency
- d) Independent of load
  **Answer: b**

17. Time-based utilization measures:

- a) Memory usage
- b) Percentage of busy time
- c) Network speed
- d) Disk space
  **Answer: b**

18. Performance bottleneck is:

- a) Fastest system component
- b) Resource limiting overall performance
- c) UI issue
- d) Syntax problem
  **Answer: b**

19. Identifying bottlenecks helps to:

- a) Optimize all components
- b) Focus optimization efforts
- c) Increase testing time
- d) Reduce documentation
  **Answer: b**

20. Performance testing should be done:

- a) Only after deployment
- b) Early and throughout SDLC
- c) Only during coding
- d) After unit testing only
  **Answer: b**

21. JMeter is:

- a) Unit testing framework
- b) Performance testing tool
- c) Security testing tool
- d) Debugger
  **Answer: b**

22. Apache JMeter is mainly used for:

- a) Functional testing only
- b) Load and performance testing
- c) UI testing
- d) Code review
  **Answer: b**

23. JMeter can test:

- a) Web applications only
- b) Web and non-web applications
- c) Desktop UI only
- d) Mobile UI only
  **Answer: b**

24. JMeter simulates:

- a) Real servers
- b) Virtual users
- c) Databases

- d) Compilers
  **Answer: b**

25. In JMeter, a Test Plan represents:

- a) Single request
- b) Entire testing scenario
- c) One assertion
- d) One thread
  **Answer: b**

26. A Thread Group in JMeter defines:

- a) Number of test plans
- b) Number of virtual users and execution
- c) Server configuration
- d) Assertions
  **Answer: b**

27. Ramp-up period in JMeter means:

- a) Time to stop test
- b) Time to start all threads
- c) Execution duration
- d) Response time
  **Answer: b**

28. Increasing ramp-up period will:

- a) Start users more gradually
- b) Start all users immediately
- c) Reduce test duration
- d) Increase response time directly
  **Answer: a**

29. Sampler in JMeter is used to:

- a) Define requests sent to server
- b) Display results

- c) Manage users
- d) Control execution order
  **Answer: a**

30. HTTP Request sampler is used to:

- a) Test database
- b) Send HTTP requests to server
- c) Analyze CPU usage
- d) Compile code
  **Answer: b**

31. Listeners in JMeter are used to:

- a) Send requests
- b) Collect and display results
- c) Manage threads
- d) Authenticate users
  **Answer: b**

32. A listener does NOT:

- a) Store results
- b) Display graphs
- c) Generate load
- d) Show statistics
  **Answer: c**

33. Assertions in JMeter are used to:

- a) Increase load
- b) Validate response content
- c) Send requests
- d) Configure server
  **Answer: b**

34. Response Assertion checks:

- a) CPU usage

- b) Response content
- c) Network speed
- d) Memory allocation
  **Answer: b**

35. Timers in JMeter are used to:

- a) Increase response time
- b) Add delays between requests
- c) Stop execution
- d) Collect results
  **Answer: b**

36. Timers help simulate:

- a) Perfect users
- b) Realistic user behavior
- c) Server failures
- d) Database crashes
  **Answer: b**

37. Controllers in JMeter are used to:

- a) Control request execution logic
- b) Display reports
- c) Store data
- d) Measure memory
  **Answer: a**

38. Logic Controllers determine:

- a) What request is sent
- b) When and how requests are executed
- c) Response validation
- d) Thread creation
  **Answer: b**

39. Pre-processors in JMeter are executed:

- a) After request
- b) Before sampler execution
- c) After listener
- d) At end of test
  **Answer: b**

40. Post-processors in JMeter are executed:

- a) Before sampler
- b) After sampler execution
- c) Before test starts
- d) After test ends
  **Answer: b**

41. Config Elements in JMeter are used to:

- a) Generate load
- b) Define default values and configurations
- c) Display results
- d) Control execution order
  **Answer: b**

42. HTTP Request Defaults is used to:

- a) Send HTTP requests
- b) Avoid repeating common HTTP configuration
- c) Display response data
- d) Validate responses
  **Answer: b**

43. CSV Data Set Config allows:

- a) Generating random users
- b) Reading test data from CSV files
- c) Recording HTTP traffic
- d) Measuring response time
  **Answer: b**

44. Using CSV Data Set Config helps in:

- a) Hardcoding test data
- b) Parameterizing test inputs
- c) Increasing compilation speed
- d) Disabling threads
  **Answer: b**

45. Parameterization in performance testing is important to:

- a) Reduce realism
- b) Simulate different users and inputs
- c) Decrease load
- d) Avoid assertions
  **Answer: b**

46. Correlation in JMeter is used to:

- a) Generate reports
- b) Handle dynamic server values
- c) Increase number of users
- d) Measure throughput
  **Answer: b**

47. Dynamic values in web applications often include:

- a) HTML tags
- b) Session IDs and tokens
- c) Static images
- d) CSS files
  **Answer: b**

48. Without correlation, a performance test may:

- a) Become more accurate
- b) Fail due to invalid dynamic data
- c) Run faster

- d) Generate more users
  **Answer: b**

49. Regular Expressions in JMeter are mainly used for:

- a) Encrypting data
- b) Extracting dynamic values
- c) Sending requests
- d) Creating threads
  **Answer: b**

50. Regular Expression Extractor is a:

- a) Sampler
- b) Listener
- c) Post-Processor
- d) Timer
  **Answer: c**

51. A Post-Processor in JMeter is executed:

- a) Before sampler
- b) After sampler
- c) Before test start
- d) After test end
  **Answer: b**

52. Distributed testing in JMeter allows:

- a) Running tests on single machine only
- b) Running load generators on multiple machines
- c) Reducing test duration only
- d) Eliminating network delays
  **Answer: b**

53. Distributed testing is mainly used when:

- a) Load is very small
- b) Single machine cannot generate required load

- c) UI testing is needed
- d) Unit testing is required
  **Answer: b**

54. In distributed testing, the controller machine is called:

- a) Slave
- b) Agent
- c) Master
- d) Worker
  **Answer: c**

55. Machines generating load in distributed testing are called:

- a) Masters
- b) Servers
- c) Slaves
- d) Controllers
  **Answer: c**

56. JMeter listeners can impact performance because they:

- a) Increase server load
- b) Consume client-side resources
- c) Reduce network latency
- d) Improve throughput
  **Answer: b**

57. It is recommended to:

- a) Use many listeners during load test
- b) Disable listeners during heavy load tests
- c) Always display graphs
- d) Ignore test results
  **Answer: b**

58. Non-GUI mode in JMeter is preferred because it:

- a) Is slower

- b) Consumes fewer resources
- c) Disables reports
- d) Limits users
  **Answer: b**

59. JMeter reports provide:

- a) Raw code
- b) Performance statistics and graphs
- c) UI mockups
- d) Security logs
  **Answer: b**

60. Average response time represents:

- a) Slowest request
- b) Fastest request
- c) Mean response time of all samples
- d) Network latency only
  **Answer: c**

61. Percentile (e.g., 90th percentile) means:

- a) 90% of requests failed
- b) 90% of responses are faster than or equal to this value
- c) Average response time
- d) Maximum response time
  **Answer: b**

62. Error percentage indicates:

- a) CPU usage
- b) Percentage of failed requests
- c) Network speed
- d) Memory leaks
  **Answer: b**

63. Throughput in JMeter reports is measured as:

- a) Requests per second/minute
- b) Seconds per request
- c) Bytes per request
- d) Users per test
  **Answer: a**

64. A spike in error rate during load test usually indicates:

- a) Improved performance
- b) System instability under load
- c) Proper scaling
- d) Reduced response time
  **Answer: b**

65. Performance baseline is:

- a) Random test result
- b) Reference performance measurement
- c) Maximum achievable throughput
- d) Worst-case response time
  **Answer: b**

66. Comparing test results against baseline helps to:

- a) Ignore regressions
- b) Detect performance degradation
- c) Reduce test scope
- d) Eliminate testing
  **Answer: b**

67. Performance regression occurs when:

- a) New version performs better
- b) Performance degrades after changes
- c) Bugs are fixed
- d) Load decreases
  **Answer: b**

68. Best practice in performance testing is to:

- a) Test only once
- b) Test with realistic data and load
- c) Ignore environment
- d) Skip monitoring
  **Answer: b**

69. Monitoring server resources during performance testing helps to:

- a) Improve UI
- b) Identify bottlenecks
- c) Reduce test duration
- d) Write test cases
  **Answer: b**

70. Performance testing results should be analyzed with respect to:

- a) UI colors
- b) Business requirements and SLAs
- c) Source code formatting
- d) Documentation size
  **Answer: b**

MCQ – Lecture 8

1. Mobile application testing focuses on:

- a) Desktop UI only
- b) Testing apps on mobile devices and environments
- c) Server-side logic only
- d) Database performance only
  **Answer: b**

2. Mobile testing is challenging mainly because of:

- a) Single operating system
- b) Limited device types
- c) Device fragmentation
- d) Static screen sizes
  **Answer: c**

3. Device fragmentation refers to:

- a) Broken devices
- b) Variety of devices, OS versions, and screen sizes
- c) Network failures
- d) Low battery
  **Answer: b**

4. Mobile applications can be classified into:

- a) Web apps only
- b) Native, Web, and Hybrid apps
- c) Desktop and web apps
- d) Games only
  **Answer: b**

5. Native mobile applications are:

- a) Accessed via browser
- b) Installed and run on device OS
- c) Platform independent

- d) Server-side only
  **Answer: b**

6. Web mobile applications are:

- a) Installed from app store
- b) Developed using native SDK
- c) Accessed through mobile browsers
- d) Hardware dependent
  **Answer: c**

7. Hybrid applications combine:

- a) Native code only
- b) Web technologies only
- c) Native and web components
- d) Desktop and server code
  **Answer: c**

8. Mobile testing should consider:

- a) CPU only
- b) Screen size and resolution
- c) Desktop keyboard
- d) Printer compatibility
  **Answer: b**

9. Screen resolution affects:

- a) Database performance
- b) UI layout and usability
- c) Network latency
- d) Server memory
  **Answer: b**

10. Orientation testing checks:

- a) CPU usage
- b) Portrait and landscape modes

- c) App installation
- d) App security
  **Answer: b**

11. Mobile usability testing focuses on:

- a) Code structure
- b) User experience and ease of use
- c) Memory leaks
- d) Encryption
  **Answer: b**

12. Mobile performance testing evaluates:

- a) UI colors
- b) App response time and resource usage
- c) Documentation quality
- d) Code readability
  **Answer: b**

13. Battery consumption testing aims to:

- a) Reduce app size
- b) Measure battery usage by app
- c) Improve UI
- d) Encrypt data
  **Answer: b**

14. Excessive battery usage may cause:

- a) Faster app execution
- b) Poor user experience
- c) Better security
- d) Lower memory usage
  **Answer: b**

15. Network testing in mobile apps focuses on:

- a) CPU load

- b) Different network conditions
- c) Code complexity
- d) UI colors
  **Answer: b**

16. Mobile apps should be tested under:

- a) One network condition
- b) Different network types (WiFi, 3G, 4G, etc.)
- c) No network
- d) Desktop network only
  **Answer: b**

17. Interrupted testing includes scenarios like:

- a) Code compilation
- b) Incoming calls or messages
- c) UI rendering
- d) Database queries
  **Answer: b**

18. Installing and uninstalling apps repeatedly tests:

- a) Memory leaks and data cleanup
- b) UI colors
- c) Code syntax
- d) Server response
  **Answer: a**

19. Compatibility testing in mobile apps ensures:

- a) Same battery usage
- b) Proper behavior across devices and OS versions
- c) Same code size
- d) Same network speed
  **Answer: b**

20. Security testing in mobile apps focuses on:

- a) UI layout
- b) Protecting user data and communication
- c) Screen resolution
- d) App size
  **Answer: b**

21. Mobile automation testing tools are used to:

- a) Replace all manual testing
- b) Automate repetitive test cases
- c) Test hardware
- d) Compile apps
  **Answer: b**

22. Emulator is:

- a) A real physical device
- b) Software that mimics mobile device behavior
- c) Network simulator only
- d) Performance tool
  **Answer: b**

23. Simulator differs from emulator because simulator:

- a) Uses real hardware
- b) Mimics OS behavior but not hardware
- c) Runs slower
- d) Tests battery usage accurately
  **Answer: b**

24. Real device testing is important because:

- a) Emulators are always accurate
- b) Some issues appear only on real devices
- c) It is cheaper
- d) It avoids automation
  **Answer: b**

25. Game testing focuses mainly on:

- a) Database correctness
- b) User experience and gameplay
- c) Compiler behavior
- d) Server installation
  **Answer: b**

26. Game testing differs from traditional testing because it:

- a) Ignores usability
- b) Emphasizes fun and playability
- c) Focuses only on code
- d) Avoids performance testing
  **Answer: b**

27. Gameplay testing checks:

- a) SQL queries
- b) Rules, logic, and flow of the game
- c) Network protocols
- d) Compiler errors
  **Answer: b**

28. Graphics testing in games focuses on:

- a) Database integrity
- b) Visual quality and rendering
- c) Encryption
- d) Memory leaks only
  **Answer: b**

29. Sound testing ensures:

- a) Background services
- b) Audio effects and synchronization
- c) Network speed

- d) UI navigation
  **Answer: b**

30. Performance testing in games focuses on:

- a) Frame rate and loading time
- b) Code readability
- c) Database normalization
- d) Documentation
  **Answer: a**

31. Frame rate (FPS) refers to:

- a) Frames per second
- b) Functions per second
- c) Files per second
- d) Frames per session
  **Answer: a**

32. Low FPS may cause:

- a) Smooth gameplay
- b) Lag and poor user experience
- c) Faster execution
- d) Better graphics
  **Answer: b**

33. Multiplayer game testing focuses on:

- a) Single device only
- b) Network latency and synchronization
- c) UI colors
- d) Code style
  **Answer: b**

34. Stress testing in games checks:

- a) Single player mode
- b) Behavior under heavy load

- c) UI layout
- d) Code comments
  **Answer: b**

35. Compatibility testing in games ensures:

- a) Same gameplay on all devices
- b) Game works on different hardware and OS
- c) Same FPS always
- d) Same resolution
  **Answer: b**

36. Localization testing in games focuses on:

- a) Graphics only
- b) Language and cultural adaptation
- c) Performance only
- d) Security
  **Answer: b**

37. Save/Load testing ensures:

- a) Game UI is correct
- b) Game progress is saved and restored correctly
- c) Network speed
- d) Code syntax
  **Answer: b**

38. Cheating prevention testing focuses on:

- a) UI layout
- b) Preventing unfair advantages
- c) Improving graphics
- d) Reducing memory usage
  **Answer: b**

39. Usability is critical in games because:

- a) Games are technical systems only

- b) Player satisfaction depends on experience
- c) Code size matters
- d) Network speed only
  **Answer: b**

40. Regression testing in mobile and games ensures:

- a) New bugs are added
- b) Existing functionality still works after changes
- c) Performance decreases
- d) UI is redesigned
  **Answer: b**

41. Crash testing in mobile applications focuses on:

- a) UI layout
- b) Application stability under unexpected conditions
- c) Database indexing
- d) Code formatting
  **Answer: b**

42. A mobile app crash is usually caused by:

- a) High usability
- b) Unhandled exceptions
- c) Good memory management
- d) Optimized code
  **Answer: b**

43. Memory testing in mobile apps aims to detect:

- a) UI bugs
- b) Memory leaks and excessive usage
- c) Network latency
- d) Screen resolution issues
  **Answer: b**

44. Mobile apps should release resources when:

- a) App is in foreground
- b) App goes to background
- c) App starts
- d) App is installed
  **Answer: b**

45. Interrupted testing also includes:

- a) Database updates
- b) Incoming notifications
- c) UI animations
- d) Code refactoring
  **Answer: b**

46. Background testing checks app behavior when:

- a) App is closed permanently
- b) App runs in background
- c) Device is powered off
- d) App is uninstalled
  **Answer: b**

47. Mobile accessibility testing focuses on:

- a) CPU utilization
- b) Supporting users with disabilities
- c) Server performance
- d) Database normalization
  **Answer: b**

48. Accessibility features include:

- a) High FPS
- b) Screen readers and voice control
- c) Encryption
- d) Compression
  **Answer: b**

49. Touch input testing verifies:

- a) Mouse behavior
- b) Gestures like tap, swipe, and pinch
- c) Keyboard shortcuts
- d) Voice commands only
  **Answer: b**

50. Gesture testing is important because:

- a) Mobile devices use keyboard primarily
- b) User interaction is gesture-based
- c) Games ignore gestures
- d) Servers handle gestures
  **Answer: b**

51. Installation testing ensures:

- a) App UI correctness
- b) App installs successfully on device
- c) App performance
- d) App security
  **Answer: b**

52. Upgrade testing checks:

- a) First installation
- b) App behavior after version update
- c) Network performance
- d) Database schema only
  **Answer: b**

53. Uninstallation testing verifies:

- a) App removes all associated data
- b) App UI responsiveness
- c) Battery usage

- d) Frame rate
  **Answer: a**

54. Push notification testing focuses on:

- a) UI layout
- b) Message delivery and handling
- c) Code syntax
- d) Compilation
  **Answer: b**

55. Offline testing checks app behavior when:

- a) Network is available
- b) Network is unavailable
- c) Server is fast
- d) CPU is idle
  **Answer: b**

56. Data synchronization testing ensures:

- a) Data consistency between client and server
- b) UI alignment
- c) Code readability
- d) Encryption strength
  **Answer: a**

57. Real device testing is essential to test:

- a) Ideal conditions only
- b) Sensors like GPS, camera, and accelerometer
- c) Code compilation
- d) Documentation
  **Answer: b**

58. Sensor testing in mobile apps includes:

- a) Database triggers
- b) GPS, camera, and gyroscope

- c) Compiler options
- d) Network routers
  **Answer: b**

59. Game balance testing focuses on:

- a) Code coverage
- b) Fairness and difficulty level
- c) Database queries
- d) Memory leaks only
  **Answer: b**

60. Game AI testing verifies:

- a) Database logic
- b) Behavior of non-player characters (NPCs)
- c) UI layout
- d) Network encryption
  **Answer: b**

61. Collision testing in games ensures:

- a) UI responsiveness
- b) Objects interact correctly
- c) Code formatting
- d) Server scalability
  **Answer: b**

62. Physics testing in games focuses on:

- a) Database performance
- b) Realistic movement and interactions
- c) Network protocols
- d) Encryption
  **Answer: b**

63. Animation testing ensures:

- a) Proper rendering and smooth transitions

- b) Database integrity
- c) Network speed
- d) Code syntax
  **Answer: a**

64. Save-point testing verifies:

- a) UI colors
- b) Correct resume from saved states
- c) CPU usage
- d) Server uptime
  **Answer: b**

65. Multiplayer synchronization issues may result in:

- a) Fair gameplay
- b) Lag or inconsistent states
- c) Faster frame rates
- d) Better graphics
  **Answer: b**

66. Network latency in online games affects:

- a) Database storage
- b) Player experience and responsiveness
- c) Code compilation
- d) UI layout
  **Answer: b**

67. Anti-cheat testing aims to:

- a) Improve graphics
- b) Prevent hacking and cheating
- c) Reduce battery usage
- d) Increase FPS
  **Answer: b**

68. Localization testing in mobile games ensures:

- a) Same language for all users
- b) Correct translation and cultural adaptation
- c) Faster execution
- d) Smaller app size
  **Answer: b**

69. Mobile game performance testing focuses on:

- a) Code readability
- b) FPS, loading time, and resource usage
- c) Database normalization
- d) Documentation
  **Answer: b**

70. Compatibility testing in mobile games is important due to:

- a) Single device type
- b) Hardware and OS fragmentation
- c) No network
- d) Desktop usage
  **Answer: b**

71. Regression testing in mobile apps ensures:

- a) New bugs are introduced
- b) Existing features still work after updates
- c) Performance always increases
- d) UI redesign
  **Answer: b**

72. Exploratory testing is useful in mobile and games because:

- a) All behavior is predictable
- b) User interactions vary widely
- c) Automation is always enough
- d) Games are static
  **Answer: b**

73. Automation testing in mobile is challenging because:

- a) No tools exist
- b) Frequent UI changes and gestures
- c) Code is static
- d) Network is stable
  **Answer: b**

74. Manual testing is still important in game testing because:

- a) Games have no logic
- b) Fun and playability are subjective
- c) Automation replaces humans
- d) Performance is irrelevant
  **Answer: b**

75. The main goal of Mobile and Game Testing is to:

- a) Eliminate all bugs
- b) Ensure good user experience and stability
- c) Reduce development cost only
- d) Increase code size
  **Answer: b**

**MCQ – Lecture 9 to slide 57**

1. A Web Service is defined as:

- a) A desktop application
- b) A software system designed to support interoperable machine-to-machine interaction
- c) A database server
- d) A UI framework
  **Answer: b**

2. Web services mainly enable:

- a) Human interaction only
- b) Application-to-application communication
- c) Hardware communication
- d) UI rendering
  **Answer: b**

3. Web services are usually accessed over:

- a) FTP
- b) HTTP/HTTPS
- c) SMTP
- d) TCP only
  **Answer: b**

4. A service-oriented architecture (SOA) is based on:

- a) Monolithic systems
- b) Loosely coupled services
- c) Single-tier architecture
- d) Tight coupling
  **Answer: b**

5. Loose coupling means:

- a) Services depend heavily on each other
- b) Services can change independently

- c) Services share databases
- d) Services run on same machine
  **Answer: b**

6. Web services are platform independent because they:

- a) Use compiled binaries
- b) Use standard protocols and formats
- c) Run on Windows only
- d) Use same programming language
  **Answer: b**

7. A client in web services is:

- a) Database
- b) Service provider
- c) Service consumer
- d) Web server
  **Answer: c**

8. A service provider is responsible for:

- a) Consuming services
- b) Publishing and hosting services
- c) Sending UI requests
- d) Testing clients
  **Answer: b**

9. Web services can communicate using:

- a) Proprietary protocols only
- b) Standard protocols
- c) Local function calls
- d) File systems
  **Answer: b**

10. The two main types of web services are:

- a) SOAP and REST

- b) JSON and XML
- c) HTTP and FTP
- d) Client and Server
  **Answer: a**

11. SOAP stands for:

- a) Simple Object Access Protocol
- b) Service Oriented Access Protocol
- c) Secure Object Access Process
- d) Simple Open API Protocol
  **Answer: a**

12. SOAP is based on:

- a) JSON only
- b) XML messaging
- c) Plain text
- d) Binary format
  **Answer: b**

13. SOAP messages are:

- a) Lightweight
- b) Stateless only
- c) XML-based and structured
- d) Browser dependent
  **Answer: c**

14. SOAP envelope contains:

- a) Header and Body
- b) URL and method
- c) Query and response
- d) Client ID only
  **Answer: a**

15. SOAP header is used for:

- a) UI layout
- b) Metadata like security and transactions
- c) Database queries
- d) Response rendering
  **Answer: b**

16. SOAP body contains:

- a) Authentication only
- b) Actual message payload
- c) Routing info
- d) HTTP headers
  **Answer: b**

17. SOAP faults are used to:

- a) Optimize performance
- b) Report errors
- c) Encrypt messages
- d) Compress data
  **Answer: b**

18. SOAP is considered:

- a) Lightweight
- b) Heavyweight compared to REST
- c) Browser-based
- d) UI-driven
  **Answer: b**

19. REST stands for:

- a) Representational State Transfer
- b) Remote Execution Standard Technique
- c) Reliable Service Transfer
- d) Response State Technology
  **Answer: a**

20. REST is an architectural:

- a) Language
- b) Protocol
- c) Style
- d) Framework
  **Answer: c**

21. RESTful services use:

- a) SOAP only
- b) XML only
- c) Standard HTTP methods
- d) FTP commands
  **Answer: c**

22. Common HTTP methods used in REST are:

- a) PUSH, POP
- b) GET, POST, PUT, DELETE
- c) SEND, RECEIVE
- d) OPEN, CLOSE
  **Answer: b**

23. GET method is used to:

- a) Create data
- b) Update data
- c) Retrieve data
- d) Delete data
  **Answer: c**

24. POST method is used to:

- a) Retrieve resource
- b) Create new resource
- c) Delete resource

- d) Cache response
  **Answer: b**

25. PUT method is mainly used to:

- a) Create or update resource
- b) Retrieve resource
- c) Read-only access
- d) Delete resource
  **Answer: a**

26. DELETE method is used to:

- a) Add data
- b) Modify data
- c) Remove resource
- d) Cache response
  **Answer: c**

27. REST is preferred over SOAP because it is:

- a) More complex
- b) Lightweight and faster
- c) XML-dependent
- d) Statefull
  **Answer: b**

28. REST services are usually:

- a) Stateless
- b) Stateful
- c) Session-based
- d) Connection-oriented
  **Answer: a**

29. Statelessness means:

- a) Server stores client session
- b) Each request contains all required information

- c) Client stores server state
- d) Server maintains history
  **Answer: b**

30. REST responses are commonly represented using:

- a) Binary files
- b) JSON or XML
- c) HTML only
- d) SQL
  **Answer: b**

31. XML stands for:

- a) Extended Markup Language
- b) Extensible Markup Language
- c) Example Markup Language
- d) External Markup Language
  **Answer: b**

32. XML is mainly used to:

- a) Execute code
- b) Store and transport data
- c) Style web pages
- d) Compile programs
  **Answer: b**

33. XML tags are:

- a) Predefined
- b) User-defined
- c) Fixed by browser
- d) Case-insensitive
  **Answer: b**

34. XML is:

- a) Case insensitive

- b) Case sensitive
- c) Language dependent
- d) Binary
  **Answer: b**

35. XML must have:

- a) Multiple root elements
- b) At least one root element
- c) No root element
- d) Optional root
  **Answer: b**

36. XML attributes provide:

- a) Data structure only
- b) Additional information about elements
- c) Encryption
- d) UI formatting
  **Answer: b**

37. Well-formed XML means:

- a) Valid against DTD only
- b) Correct syntax rules followed
- c) Optimized structure
- d) Browser compatible
  **Answer: b**

38. Valid XML means:

- a) Syntax is correct
- b) Conforms to DTD or schema
- c) Has comments only
- d) Uses JSON
  **Answer: b**

39. JSON stands for:

- a) Java Syntax Object Notation
- b) JavaScript Object Notation
- c) Java Standard Object Network
- d) Joint System Object Notation
  **Answer: b**

40. JSON is:

- a) Binary format
- b) Text-based data format
- c) Markup language
- d) Database language
  **Answer: b**

41. JSON is preferred because it is:

- a) Verbose
- b) Lightweight and easy to parse
- c) XML-dependent
- d) Case-insensitive
  **Answer: b**

42. JSON represents data using:

- a) Tags
- b) Key-value pairs
- c) Tables
- d) Attributes only
  **Answer: b**

43. JSON arrays are enclosed using:

- a) {}
- b) <>
- c) []
- d) ()
  **Answer: c**

44. JSON objects are enclosed using:

- a) []
- b) <>
- c) {}
- d) ()
  **Answer: c**

45. JSON supports data types like:

- a) int, float only
- b) string, number, boolean, array, object
- c) class, interface
- d) pointer
  **Answer: b**

46. Comparing XML and JSON, JSON is:

- a) More verbose
- b) Heavier
- c) Easier to read and write
- d) Schema dependent
  **Answer: c**

47. Web service testing focuses on:

- a) UI look and feel
- b) Validating requests and responses
- c) Screen resolution
- d) Animations
  **Answer: b**

48. API testing is usually done:

- a) After UI testing
- b) Independently of UI
- c) By end users

- d) Without automation
  **Answer: b**

49. Web service testing checks:

- a) Only functionality
- b) Functionality, performance, and security
- c) UI layout
- d) Compilation errors
  **Answer: b**

50. Request validation ensures:

- a) Server crashes
- b) Correct format and parameters
- c) UI correctness
- d) Network speed
  **Answer: b**

51. Response validation checks:

- a) Server configuration
- b) Status code and response body
- c) Database schema
- d) UI navigation
  **Answer: b**

52. HTTP status code 200 indicates:

- a) Error
- b) Redirect
- c) Success
- d) Unauthorized
  **Answer: c**

53. HTTP status code 400 indicates:

- a) Success
- b) Client error

- c) Server error
- d) Redirection
  **Answer: b**

54. HTTP status code 500 indicates:

- a) Client error
- b) Server error
- c) Redirect
- d) Success
  **Answer: b**

55. API testing tools include:

- a) Selenium only
- b) Postman
- c) Photoshop
- d) Eclipse
  **Answer: b**

56. Postman is mainly used to:

- a) Design UI
- b) Send and test API requests
- c) Compile code
- d) Monitor CPU
  **Answer: b**

57. A successful web service test ensures:

- a) UI attractiveness
- b) Correct, secure, and reliable service behavior
- c) Smaller code size
- d) Fewer users

**Answer: b**